Yashwanth Varma Vadlamudi

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EDUCATION

Cornell University **Ithaca, NY**

Master of Engineering in Computer Science *Jan 2025 – Dec 2025*

– **Relevant Courses:** Distributed Computing Principles, Database Management Systems, System Security, Machine Learning, Computer Networks, System Modelling with Generative AI.

Manipal University Jaipur **Jaipur, India**

Bachelor of Technology in Computer Science and Engineering *July 2019 – May 2023*

WORK EXPERIENCE

Cornell University **Ithaca, NY**

Research Fellow *Feb 2026 – Present*

– Built a GATv2/LSTM traffic prediction engine in Python modeling road networks as dynamic spatial graphs across 20 U.S. cities; achieved R^2 of 0.98 with 16-city incremental training.
– Engineered a multi-agent LLM pipeline using AutoGen and LiteLLM to simulate urban mobility patterns with structured JSON validation to reduce hallucinations.
– Built an end-to-end emissions analyzer using React, Node.js, and PostgreSQL on AWS; designed a FastAPI orchestrator and horizontally scaled Docker architecture across EC2 instances, reducing runtime by 94%.

Cimpress

Software Engineer *Aug 2023 – Jan 2025*

– Architected a highly scalable discount-matching REST API to return discounts applicable to the cart in Node.js using Abstract Syntax Trees and GraphQL, processing 1M+ monthly cart transactions.
– Re-architected the core cart recalculation service in Node.js and TypeScript to handle high-throughput traffic, reducing checkout latency by 25% through asynchronous API parallelization and intelligent payload caching.
– Resolved 10+ critical API errors, eliminating order blockages to improve checkout reliability for 6,000+ monthly users.
– Engineered an interactive discount UX using React, Hooks, and Redux for complex state management, directly driving a 25% increase in average order value (AOV).
– Awarded High Ownership Award for technical excellence, proactive system analysis, and clear documentation.

Software Engineer Intern *Feb 2023 – Aug 2023*

– Optimized React performance by serving translation files as chunks via AWS S3, reducing the bundle size by 75%.
– Contributed to a major website rebranding by developing a suite of reusable React components, ensuring cross-platform consistency with the updated brand identity and design system.

PUBLICATIONS

MR images-based Alzheimer's Detection using AlexNet and InceptionV3 (2024) | *IEEE Conference*
Windows Malware Detection Using CNN and AlexNet Learning Models (2022) | *Springer Conference*

ACADEMIC PROJECTS

Distributed, Fault-Tolerant Key-Value Store | *Java, Paxos, Two Phase Commit (2PC)*

– Architected a custom Multi-Paxos sharded key-value store using hash-based sharding for distributed data consistency.
– Integrated a two-phase commit (2PC) protocol to support atomic multi-key transactions and strict fault tolerance.

TDM Activity Generation Pipeline | *Python, AWS Bedrock, Lambda, Fargate, DynamoDB*

– Built a serverless AWS pipeline using Bedrock batch inference to generate realistic daily schedules for 100k+ synthetic urban personas, powering travel demand modeling research.
– Redesigned memory generation step from sequential to 10-shard parallel Fargate architecture, achieving 10x speedup (63 min → 25 min) and unblocking multi-day city-scale simulations.

SKILLS

Languages: Java, Go, Rust, Python, TypeScript, JavaScript, C++, C#, SQL, Shell
Full-Stack & Web: React, Redux, HTML5/CSS3, Node.js, GraphQL, REST, API Design, State Management
Distributed Systems: Microservices, gRPC, Kafka, Consensus Algorithms (Paxos, Raft), 2PC
AI & Machine Learning: Generative AI, PyTorch, LLM Inference Pipelines, AI Coding (Cursor, Copilot)
Cloud & Databases: AWS (EKS, ECR, EC2, S3, Lambda), Docker, Kubernetes, Terraform, CI/CD, PostgreSQL, MongoDB, DynamoDB, Redis